Dart Programming Language

1. If Statement

```
void main() {
  int age = 25; // 15
  if (age >= 18) {
    print("Adult");      // Adult
  }
}
```

```
void main() {
  bool isloggedIn = true;
  if (isLoggedIn) {
    print('Welcome Back');
    print('Loading your dashboard...');
    print("You have 3 notifications');
  }
}
```

2. If-Else Statement (Two Way Branching)

```
void main() {
  int marks = 30; // 50
  if (marks >= 33) {
    print('Pass');      // Pass
  } else {
    print('Fail');      // Fail
  }
}
```

3. Else-if Ladder

```
void main() {
  int score = 25; // 54
  if (score >= 90) {
    print('A Grade');
  } else if (score >= 75) {
    print('B Grade');
  }
```

```
else if (score >= 60) {
print('C Grade');
} else if (score >= 33) {
print('D Grade');
} else {
print('Fail');
}
}
```

4. Logical operator

```
void main() {
int age = 19;
bool hasLicence = true;
// AND operator
if (age >= 18 && hasLicence) {
print('Can drive');
}
// OR operator
bool isWeekend = true;
bool isHoliday = false;
if (isWeekend || isHoliday) {
print('No Work Today');
}
// NOT operator
bool isLoggedIn = false;
if (! isLoggedIn) {
print('Please Login');
}
}
```

5. Nested if Statement (Hierarchical Condition)

```
void main() {
int age = 18;
bool hasID = false;
```

```
if(age >= 18) {
print("Age check passed");
if (hasID) {
print("ID check passed");
print("Allowed to enter");
} else {
print('ID required');
}
} else {
print ('Too young');
}
}
```

6. Ternary Operator

```
void main () {
int age = 17;
String status = age >= 18 ? "Adult" : "Minor";
print (status);
// Finding Maximum
int a = 10, b = 20;
int max = a > b ? a : b;
print (max);
// Nested Ternary
int score = 65;
String grade = score >= 90
? "A"
: score >= 80
? "B"
```

: score >= 70
? 'C'
: 'F';
print('Grade : $grade'); // F
}

7. Switch Statement (Old)

```
void main () {
int day = 9;
switch (day) {
  case 1:
    print('Monday');
  break;
  case 2:
    print('Tuesday');
  break;
  case 3:
    print('Wednesday');
  break;
  default:
    print('Invalid Day');
  }
}
```

Switch Expression (Dart 3)

```
void main () {
String day = "Sunday";
String message = switch (day) {
  'Monday' => 'Working Day', (1)
  'Sunday' => 'Holiday', (2)
  _ => 'Normal Day'
};
```

```
print (message);
}
```

8. if case Statement (Dart 3)

```
void main () {
var point = [10, 20];
if (point case [int x, int y]) {
print ('x = $x');
print ('y = $y');
}
}
```

x = 10
y = 20

9. Pattern Matching

```
void main () {
var data = (10, 20);
switch (data) {
case (int x, int y) :
print ("x = $x, y = $y");
}
}
```

// x = 10, y = 20

10. null coalescing operator

```
void main () {
String? name;
String userName = name ?? "Guest";
print (userName);
}
```

// Guest

Home Work

1. write a program that checks
   - Positive Number
   - Negative Number
   - Zero
2. Take a weekday name and print
   - Working Day
   - Weekend

   using switch
3. Ternary op (Even / odd)